



Introduction to Linux OS

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Linux File Permissions

In Linux, file permissions control who can **read**, **write**, or **execute** a file or directory. These permissions help maintain system security and prevent unauthorized access.

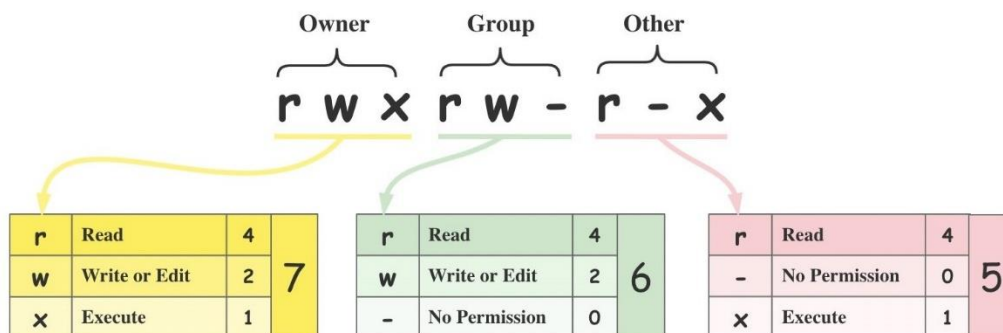
```
$ ls -l
total 12
-rw-r--r-- 1 centos9 centos9 56 Aug 5 16:52 file.py
drwxr-xr-x 2 centos9 centos9 4096 Aug 5 16:51 Games
-rw-r--r-- 1 centos9 centos9 56 Aug 5 16:52 hello.txt
```

-/d **rw-r--r--** **1** **centos9** **centos9** **56** **Aug 5** **16:52** **file.py**

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Type (file/dir) User Group Other No. hard links Owner/User Group Size Date Time Name

Binary	Octal	String Representation	Permissions
000	0 (0+0+0)	---	No Permission
001	1 (0+0+1)	--x	Execute
010	2 (0+2+0)	-w-	Write
011	3 (0+2+1)	-wx	Write + Execute
100	4 (4+0+0)	r--	Read
101	5 (4+0+1)	r-x	Read + Execute
110	6 (4+2+0)	rw-	Read + Write
111	7 (4+2+1)	rwX	Read + Write + Execute





Manipulating Files

Change mode of a file

```
$ chmod [u/g/o/a] [+/-/=] [r/w/x] [filename/dirname]
```

Type	Explanation
u	user
g	group
o	other
a	all three

Type	Explanation
+	add permission
-	remove permission
=	set only this permission to u/g/o

Use **-R** flag to apply the permission changes to all files and directories within a specific directory.

Function	Example
Adding Permissions	<pre>\$ chmod a+w file.py \$./file.py \$ chmod u+x file.py \$ chmod -R a+w Games/</pre>
Removing Permissions	<pre>\$ chmod -R a-w Games/ \$ chmod -R a=w Games/ \$ chmod a=wr file.py</pre>
Adding permissions to all (user, group, other users)	<pre>\$ chmod +x file.txt \$ chmod g+w,o-rw,a+x file.txt</pre> » Use comma-separated permissions <u>without spaces</u>.

Changing the owner of the file

```
$ chown Options New_Owner:NEW_GROUP [filename/dirname]
```

Examples
<pre>\$ chown Ahmed file.txt \$ chown :developers file.txt \$ chown Ahmed:developers file.txt \$ chown -R Ahmed Games</pre>



Text Editors (Nano, Vi/Vim)

1) Nano

```
$ nano file.txt
```

If the file exists, nano will open it, otherwise it will create a new file.

Description	Command
Help	CTRL + G
Save	CTRL + S
Save As	CTRL + O
Exit/Close	CTRL + X
Highlight/Select	CTRL + 6
Copy	ALT + 6
Cut	CTRL + K
Paste	CTRL + U
Redo	ALT + E
Undo	ALT + U





2) Vi/Vim

`$ vi file.txt` **or** `$ vim file.txt`

Mode	Key	Description
Command Mode	ESC	- The default mode when you start Vi/Vim. - You use this mode to execute commands for navigating, copying, deleting, and more.
Insert Mode	i	- This is where you can type and edit text. - Press ESC to return to Command Mode.
Visual Mode	v	- Used for highlighting/selecting text. - Press ESC to return to Command Mode.

```
MG@centos9:~/Downloads - vim file1.txt
Hello World

"file1.txt" 1L, 12B                      1,11                      All
```



Function	Key	Mode	Description
Copy (Yank)	y	Visual/Command	Copy the selected text or a specified range.
Paste Text	p	Command Mode	Paste the copied text after the cursor.
Delete Text	d	Visual/Command	Delete the selected text or a specified range.
Delete Line	dd or [num]dd	Command Mode	Delete the current line(s).
Delete Word	dw or d[num]w	Command Mode	Delete a word(s).
Undo Change	u	Command Mode	Undo the last change.
Redo Change	CTRL + r	Command Mode	Redo the change that was undone.
Searching	/<pattern> or ?<pattern>	Command Mode	Start a forward/reverse search for <pattern>.
Move to Line Start	0	Command Mode	Move the cursor to the beginning of the current line.
Move to Line End	CTRL + e	Command Mode	Move the cursor to the end of the current line.
Save and Exit	:wq or ZZ	Command Mode	Save changes and exit Vim.
Exit Without Saving	:q!	Command Mode	Exit Vim without saving changes.
Split Window (Horizontal)	:split <filename>	Command Mode	Open a file in a new horizontal split window.
Split Window (Vertical)	:vsplit <filename>	Command Mode	Open a file in a new vertical split window.
Move Between Splits	CTRL + w + arrow key or h, j, k, l	Command Mode	Move between split wind

Graphical Editors

So far we've learned about various command line editors. However, if you are running Linux as a desktop operating system you might be interested in some graphical text editors and word processors.

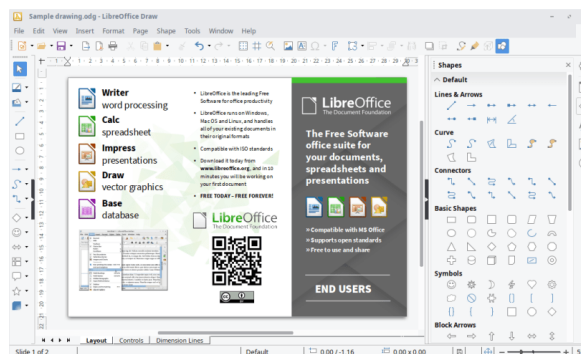
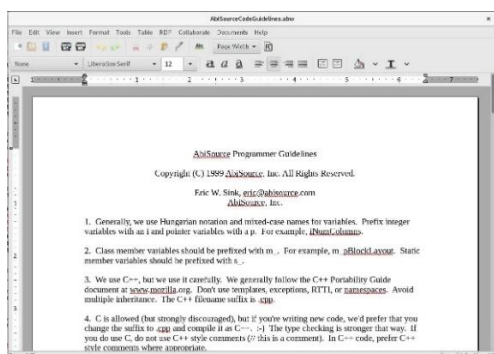
- » **emacs** – Emacs has both graphical and command line modes.
- » **gedit** – The default text editor for the Gnome desktop environment.
- » **gvim** – The graphical version of vim.
- » **kedit** – The default text editor for the [KDE \(K Desktop Environment\)](#) desktop environment.



Microsoft Word Replacement

If you are looking for a Microsoft Word replacement, consider

- [AbiWord](#) or
- [LibreOffice](#) which is a complete office suite with a spreadsheet program, a database, and presentation software.



Code/Text Editors for Linux

- [Visual Studio Code \(VS Code\)](#)
- [Sublime Text](#)
- Gedit – GNOME's simple and lightweight text editor.
- Kate
- etc.

Steps to Set Your Default Editor

You can set `$EDITOR` in your personal initialization files to ensure your favorite editor is used, be it **nano**, **emacs**, **vi/vim**, or something else.

- 1) Open your shell's configuration file (i.e., `~/.bashrc`)

```
$ nano ~/.bashrc
```

- 2) Set the `$EDITOR` variable

```
$ export EDITOR='vim'
```

- 3) Apply the changes

```
$ source ~/.bashrc
```

```
$ echo $EDITOR
```

Now, whenever you invoke a command that requires an editor (e.g., `crontab -e`, `git commit`, etc.), it will open in your preferred editor.